

Asher Straus

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EDUCATION

Northwestern University | McCormick School of Engineering | Evanston, IL Sept 2023 – Jun 2027
B.S. Manufacturing & Design Engineering | *Minor in Art* | *Segal Design Certificate* | *Robotics Certificate*
GPA: 3.6 / 4.0 Cumulative · **3.9 / 4.0 since sophomore year**

WORK EXPERIENCE

Siemens Healthineers Jun 2026 – Dec 2026

Mechanical Engineering Intern, Molecular Imaging · Hoffman Estates, IL

- Designing SPECT and PET scanner assemblies with focus on structural reliability, tolerance stack-up analysis, and DFM.
- Collaborating with the Molecular Imaging team to advance scanner hardware from early concept through design validation.

MIT Beaverworks Summer Institute

Jan 2024 – Feb 2024

Teaching Assistant: "Basics of ASICs" · Cambridge, MA

- Guided students through ASIC schematic design, simulation, and testing; collaborated with instructors to refine labs and curriculum.

P1 Motor Club

Apr 2023 – Present

Promotional Representative, Research Assistant & Test Driver Intern · Tampa, FL

- Tested new track layouts in simulators and delivered performance feedback to engineers, ensuring data-driven design improvements.
- Fabricating a miniature scale model of the racetrack as a sales tool; grew brand visibility at prospective-member events.

Monticello Motor Club

Jun 2018 – Jul 2022

Go-Kart Instructor, Technician & Social Media Coordinator · Monticello, NY

- Maintained and repaired kart systems — diagnosing mechanical faults and applying technical fixes for safe, reliable client operation.
- Instructed drivers at all skill levels with tailored coaching methods, improving safety outcomes and customer satisfaction.

PROJECT & DESIGN EXPERIENCE

Smart Sheet Smith — Advanced Intelligent Manufacturing Lab (NSF ERC-HAMMER) Sept 2025 – Jun 2026

Undergraduate Researcher · Co-author, MSEC 2026 / NAMRC54

- Co-authored a published paper on a five-agent AI pipeline that converts a 2D sheet-metal drawing into a verified, physics-grounded bending process plan — no 3D CAD model or labeled training data required.
- Built the Tracer Agent's evaluation corpus end to end — CAD parts, multi-view drawings, and hand-derived ground-truth bend math — and applied press-brake manufacturability checks the math alone cannot catch.
- Automated dataset creation with a SolidWorks VBA pipeline that randomizes base-model dimensions to export hundreds of valid, fully-dimensioned drawings — cutting data generation from days to minutes.

Formula SAE Brake System — Northwestern Formula Racing

Sep 2024 – Sep 2025

Brake System Design, Analysis & Fabrication Lead

- Designed and validated the FSAE braking system for 1G deceleration (~690 lbf tire force from ~42 lbf pedal — a ~16× gain).
- Built an analytical brake model and improved the team MATLAB sim to tune rotor size, MC bore, pedal ratio, and brake bias.
- Verified strength and thermal limits via ANSYS FEA; machined and lathe-finished rotors, routed lines, and bled the system on-car.

CT Scanner — Major Aquarium, Chicago, IL

Sept 2025 – Present

Design & Build Engineer

- Adapting a medical CT scanner for aquatic animals alongside a veterinary doctor — building stabilization and ventilation systems.
- Developing corrosion-resistant protections to shield scanner components from saltwater exposure and extend operational lifespan.

Bionic Wrench Assembly Fixture — DSGN 386, Northwestern

Mar 2025 – Jun 2025

Designer & Manufacturing Engineer

- Iterated 7 fixture versions: 3D-printed body, steel locating pins, angled stand, waterjet force plate, and a riveter-track guide.
- Built the SOP, value stream map, and line balancing analysis for a 1M-unit/year plan; pilot trial averaged a 3:58 cycle time.

ExtendIt — Desk Extension for Lecture Halls, Northwestern

Jan 2025 – Mar 2025

Designer & Engineer

- Prototyped a locking desk extension expanding lecture-hall surface area by 50%; ran 22 user interviews and 7 mockups, stress-tested to 30+ lb, and modeled cost-effective scale-up.

Assistive Devices for Adults with Cerebral Palsy — Misericordia Heart of Mercy

Jan 2024 – Jul 2024

Project Manager & Fabrication Lead · Evanston, IL

- StimSpin: built a spinning-wheel device with tactile, visual, and auditory stimulation for limited-mobility residents.
- The Pillowscape: co-designed a customizable patchwork pillow system to reduce sensory overstimulation, tested with users.

RELEVANT COURSES & CLUBS

Courses: Manufacturing Engineering Design (DFM); Computer Integrated Manufacturing; Geometry & Manufacturing; Mechanical Design & Manufacturing; Materials Selection; Mechatronics; Mechanics of Materials; Electronics Design; Fundamentals of Circuits; Human-Centered Product Design; Optimization; Python

Clubs: Formula SAE; Combat Robotics; Cookology

TECHNICAL SKILLS

Fabrication & Hardware: Circuit Design, Radio Control, 3D Printing, Welding, Machining (Lathe, CNC, Milling, Drill Press), Rapid Prototyping, Lab Testing, Design for Reliability, Design for Cost, Design for Manufacturing

Software: SolidWorks/CAD, Ansys, MATLAB, Python, Unity, Adobe Photoshop, FEA & Simulation Tools, GitHub, Visual Studio